

HighamBench N/L Results

T1–T2 from r11; T3 from combined r09–r10

■ N: no NumStability ■ L: NumStability available

Each task value is the arithmetic mean of three repetitions. Each tier mean gives every completed task equal weight. Time ends at the authenticated final submission; hidden validation is excluded. Physical LOC counts only the submitted source; imported Mathlib and NumStability implementation lines are not counted.

Tiers. T1 needs one substantial NumStability result; T2 needs at least two; T3 is at least as difficult as T2 but has no corresponding NumStability result expected to help L.

Isolation. Task attempts are isolated from one another: each uses fresh agent and workspace state and sees only its own task context, never another task’s prompt, conversation, search trace, or solution.

Coverage and success

Tier	Source	Tasks in mean	N success	L success
T1	r11	12/12	36/36	36/36
T2	r11	3/3	9/9	9/9
T3	combined r09–r10	23/23	69/69	69/69

Equal-weight tier means

Tier	Time (s)		Tokens [†]		Physical LOC	
	N	L	N	L	N	L
T1	764.9	368.8	1.259M	0.959M	302.2	182.1
T2	1531.1	465.2	3.672M	1.149M	677.3	269.1
T3	645.5	601.7	2.917M	2.922M	414.3	432.4

[†]All 228 selected attempt token records are incomplete. Token values are provider-observed lower bounds, not exact total spending.

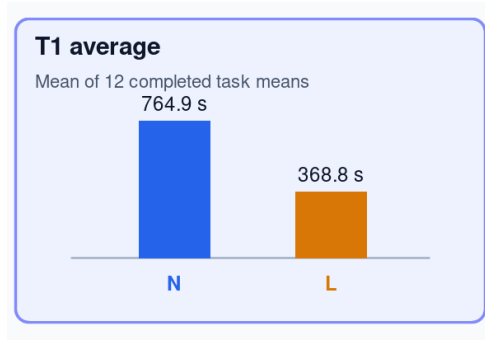
Result summary. For T1 and T2, L has lower mean elapsed time, observed-token lower bound, and submitted physical LOC. For T3, the tier means are close across conditions and show no consistent NumStability advantage. The task plots below retain their original per-task scales, so bar heights should be compared only within a panel.

Faster time, more tokens. Some T1 L runs finish sooner yet use more observed tokens because NumStability search and read outputs enlarge the context processed again on later model calls, including cached input, while theorem reuse shortens proof development and Lean checking.

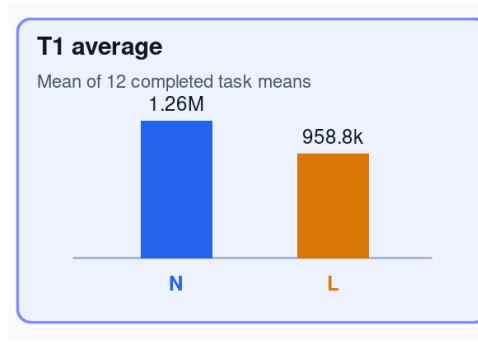
Tier mean diagrams

Each panel reports the separate absolute N and L tier means. Completed tasks receive equal weight.

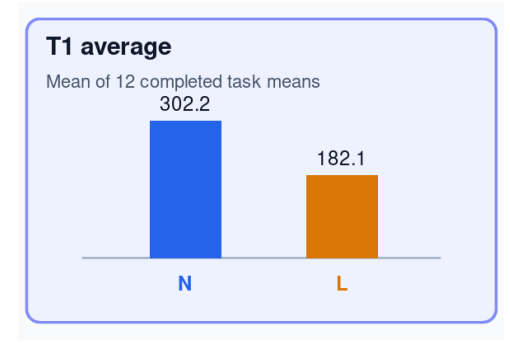
Tier 1 — 12 completed task means



Elapsed time

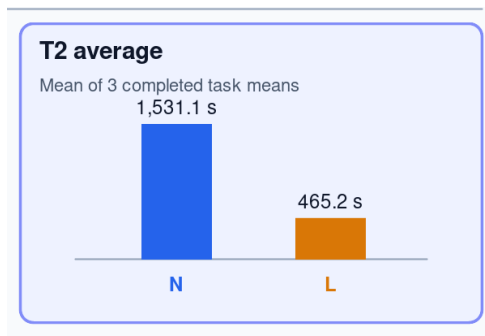


Observed tokens (lower bound)

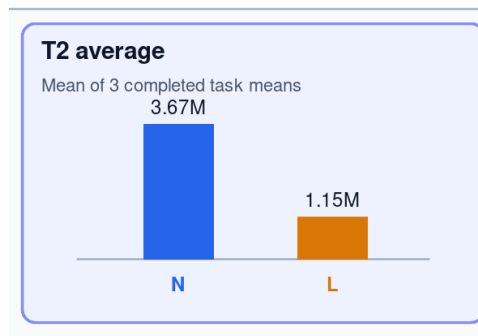


Final source lines

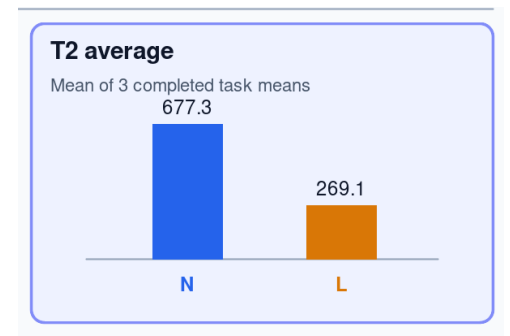
Tier 2 — 3 completed task means



Elapsed time

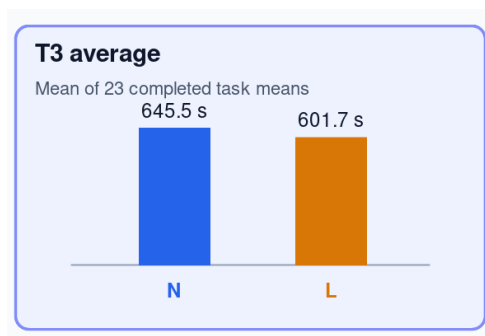


Observed tokens (lower bound)

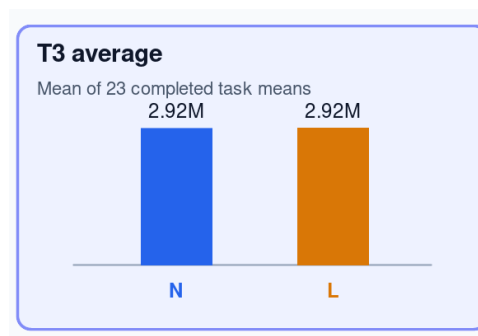


Final source lines

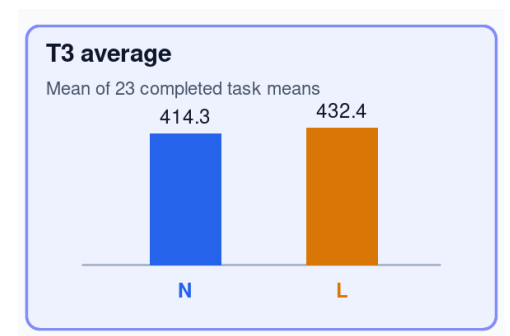
Tier 3 — 23 completed task means



Elapsed time



Observed tokens (lower bound)

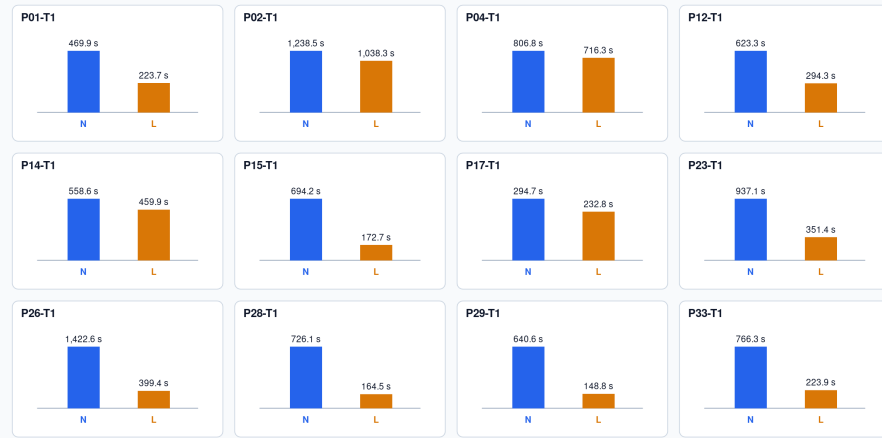


Final source lines

Tier 1 — r11

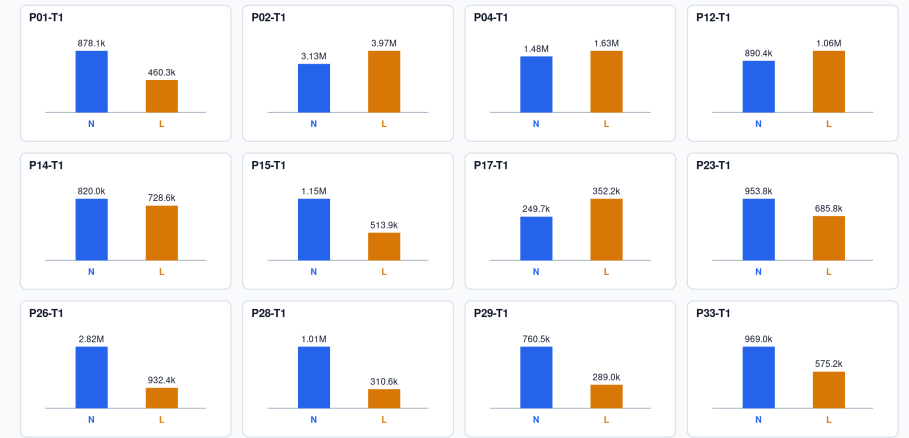
Twelve tasks, each with three accepted N/L pairs. Absolute three-run task means are shown below.

T1 — 12/12 three-run averages ready



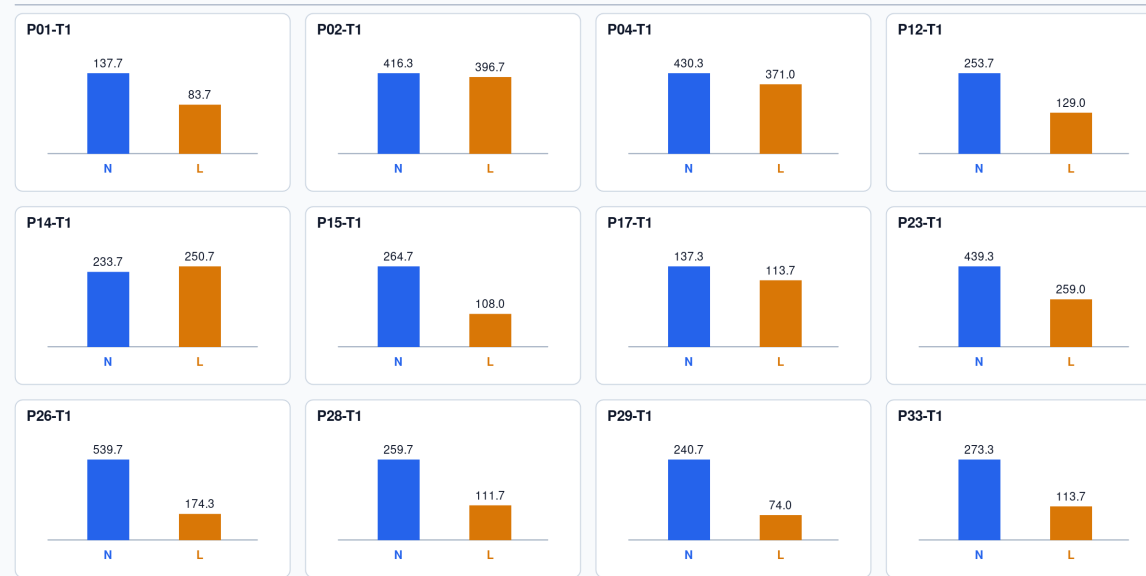
Elapsed time

T1 — 12/12 three-run averages ready



Observed tokens; all values are lower bounds

T1 — 12/12 three-run averages ready



Final submitted source lines

Tier 2 — r11

Three tasks, each with three accepted N/L pairs. Absolute three-run task means are shown below.



Tier 3 — combined r09–r10: time and tokens

Twenty-three tasks, each with three accepted N/L pairs. Absolute three-run task means are shown below.



Tier 3 — combined r09–r10: physical LOC

LOC counts only submitted candidate source; imported module internals are excluded.



Final submitted source lines